

✓ Bases and Dimension

Def! Let V be a vector space over F . A subset S of V is said to be linearly dependent (or, dependent) if $\exists$ distinct vectors $\bar{\alpha}_1, \bar{\alpha}_2, \dots, \bar{\alpha}_n \in S$ and scalars $c_1, c_2, \dots, c_n \in F$, not all of which are $\bar{0}$, s.t.
$$c_1 \bar{\alpha}_1 + c_2 \bar{\alpha}_2 + \dots + c_n \bar{\alpha}_n = \bar{0}.$$

A set which is not linearly dependent is called linearly independent. If the set S contains only finitely many $\bar{\alpha}_1, \bar{\alpha}_2, \dots, \bar{\alpha}_n \in V$, we sometimes say that $\bar{\alpha}_1, \bar{\alpha}_2, \dots, \bar{\alpha}_n$ are dependent (or independent) instead of saying S is dependent (or independent).

The following are easy consequences:

1. Any set which contains a linearly dependent set is linearly dependent.
2. Any subset of linearly independent set is linearly independent.
3. Any set which contains the $\bar{0}$ is linearly dependent.

Example: P be an $n \times n$ invertible matrix over F .
 $P_1, \dots, P_n$, the columns of P , form a basis for
the space of column matrices, $F^{n \times 1}$. If X
is a column matrix, $PX = x_1 P_1 + \dots + x_n P_n$.

$\therefore PX = 0$ has only the trivial solⁿ? $X = 0$,

$\{P_1, \dots, P_n\}$ is a linearly independent set.
Spans $F^{n \times 1}$: let Y be a column matrix. If

$X = P^{-1}Y$, then $Y = PX$, i.e.,

$$Y = x_1 P_1 + \dots + x_n P_n.$$

So, $\{P_1, \dots, P_n\}$ is a basis for $F^{n \times 1}$.

Example: let F be a subfield of $\mathbb{C}$.
 V be the space of polynomial $f(x)$ over F ;

$$f(x) = c_0 + c_1 x + \dots + c_n x^n.$$

let $f_k(x) = x^k$, $k = 0, 1, 2, \dots$. The (infinite) set
 $\{f_0, f_1, f_2, \dots\}$ is a basis for V . Clearly,
the set spans V , because the f_k (above)

$$\text{is } f = c_0 f_0 + c_1 f_1 + \dots + c_n f_n.$$

Thm. Let V be a vector space which is
spanned by a finite set of vectors $\bar{\beta}_1, \bar{\beta}_2, \dots, \bar{\beta}_m$.
Then any independent set of vectors in V is
finite and contains no more than m elements.

Coordinates

A basis B in an n -dim space V enables introduction of coordinates in V analogous to the 'natural coordinates' x_i of a $\bar{x} = (x_1, \dots, x_n) \in F^n$. Then the coordinates of $\bar{x} \in V$ relative to B will be scalars which serve to express $\bar{x}$ as a linear combination of the vectors in the basis. The natural coordinates of $\bar{x} \in F^n$ is defined by $\bar{x}$ and the std basis for F^n . If $\bar{x} = (x_1, \dots, x_n) = \sum x_i e_i$ and B is the std. basis for F^n , how are the coordinates of $\bar{x}$ determined by B & $\bar{x}$?

Defⁿ: If V is a finite-dimensional vector space, an ordered basis for V is a finite sequence of vectors which is linearly independent and spans V .

If the sequence $\bar{\alpha}_1, \bar{\alpha}_2, \dots, \bar{\alpha}_n$ is an ordered basis for V , then the set $\{\bar{\alpha}_1, \dots, \bar{\alpha}_n\}$ is ~~the~~ a basis for V . The ordered basis is the set, together w/ the specified ordering. W/ slight abuse of notation: $B = \{\bar{\alpha}_1, \dots, \bar{\alpha}_n\}$ is an ordered basis for V .

✓ V is a finite-dim vector space over F and $B = \{\bar{\alpha}_1, \dots, \bar{\alpha}_n\}$ is an ordered basis for V . For $\bar{\alpha} \in V$, $\bar{\alpha} = \sum_{i=1}^n x_i \bar{\alpha}_i$ for some unique n -tuple $(x_1, x_2, \dots, x_n)$. It's unique because if $\bar{\alpha} = \sum_{i=1}^n y_i \bar{\alpha}_i$, then

$$\bar{\alpha} - \bar{\alpha} = \sum_{i=1}^n (x_i - y_i) \bar{\alpha}_i = \bar{0} \Rightarrow x_i = y_i \forall i.$$

x_i is called i^{th} coordinate of $\bar{\alpha}$ relative to an ordered basis B .

If $\bar{\alpha} = \sum x_i \bar{\alpha}_i$ & $\bar{\beta} = \sum y_i \bar{\alpha}_i$,
then $\bar{\alpha} + \bar{\beta} = \sum (x_i + y_i) \bar{\alpha}_i$ ^{of $\bar{\alpha} + \bar{\beta}$}
 i^{th} coordinate w.r.t B .

i^{th} coordinate of $c\bar{\alpha}$ is $c\bar{\alpha}_i$ w.r.t. $\mathcal{B}$.

Note that every n -tuple $(x_1, \dots, x_n) \in F^n$ is the n -tuple of coordinates of some vector in V , namely the vector $\sum_{i=1}^n x_i \bar{\alpha}_i$.

I.e., each ordered basis for V determines a one-to-one correspondence

$$\bar{\alpha} \longrightarrow (x_1, \dots, x_n)$$

b/w the set of all vectors in V & the set of all n -tuples in F^n . This correspondence has the property that the correspondent of $(\bar{\alpha} + \bar{\beta})$ is the sum in F^n of the correspondents of $\bar{\alpha}$ & $\bar{\beta}$, and that the correspondent of $c\bar{\alpha}$ is the product in F^n of the scalar c & the correspondent of $\bar{\alpha}$.

Coordinate matrix of $\bar{\alpha}$ w.r.t. the ordered basis $\mathcal{B}$, $[\bar{\alpha}]_{\mathcal{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$.

$$\bar{\alpha} = \sum_{i=1}^n x_i \bar{\alpha}_i$$

Suppose V is n -dim. and $B = \{\bar{\alpha}_1, \bar{\alpha}_2, \dots, \bar{\alpha}_n\}$ and $B' = \{\bar{\alpha}'_1, \dots, \bar{\alpha}'_n\}$ are two ordered bases for V . There are unique scalars P_{ij} s.t.

$$\bar{\alpha}'_j = \sum_{i=1}^n P_{ij} \bar{\alpha}_i, \quad 1 \leq j \leq n.$$

Let $x'_1, \dots, x'_n$ be the coordinates of a given $\bar{\alpha}$ in the ordered basis B' . Then

$$\bar{\alpha} = x'_1 \bar{\alpha}'_1 + \dots + x'_n \bar{\alpha}'_n$$

$$= \sum_{j=1}^n x'_j \bar{\alpha}'_j$$

$$= \sum_{j=1}^n x'_j \sum_{i=1}^n P_{ij} \bar{\alpha}_i$$

$$= \sum_{i=1}^n \left(\sum_{j=1}^n (P_{ij} x'_j) \right) \bar{\alpha}_i$$

$$\bar{\alpha} = \sum_i \left(\sum_j P_{ij} x'_j \right) \bar{\alpha}_i$$

Then $\bar{\alpha} = \sum_i x_i \bar{\alpha}_i$, $(x_1, \dots, x_n)$ being coordinates w.r.t. B ,

$$x_i = \sum_j P_{ij} x'_j, \quad 1 \leq i \leq n.$$

Let $P_{n \times n}$ whose i, j entry is the scalar P_{ij} ,
and let X and X' be the coordinate
matrices of the vector $\bar{\alpha} \in V$ in the ordered
bases B & B' . Then,

$$x_i = \sum_{j=1}^n P_{ij} x'_j \quad \forall i \in \{1, \dots, n\}$$

can be expressed as $X = PX'$.

Since B & B' are linearly independent
sets, $X=0$ iff $X'=0$. This implies,
 P is invertible. Hence, $X' = P^{-1}X$.

I.e.,

$$[\bar{\alpha}]_B = P [\bar{\alpha}]_{B'}$$

$$[\bar{\alpha}]_{B'} = P^{-1} [\bar{\alpha}]_B$$

Thm. Let V be an n -dim vector space
over F . Let B & B' be two ~~ordered bases~~
of V . Then there is a unique, necessarily
invertible, $n \times n$ matrix P over F s.t.

(i) $[\bar{\alpha}]_B = P [\bar{\alpha}]_{B'}$

(ii) $[\bar{\alpha}]_{B'} = P^{-1} [\bar{\alpha}]_B$

$\forall \bar{\alpha} \in V$. Columns P_j of P are $P_j = [\bar{\alpha}_j']_B$,
for $j \in \{1, \dots, n\}$.

Thm. Suppose P is an $n \times n$ invertible matrix over F . Let V be an n -dim vector space over F , and let $\mathcal{B}$ be an ordered basis of V . Then there is a unique ordered basis $\mathcal{B}'$ of V s.t.

$$\textcircled{1} [\bar{\alpha}]_{\mathcal{B}} = P [\bar{\alpha}]_{\mathcal{B}'}$$

$$\textcircled{10} [\bar{\alpha}]_{\mathcal{B}'} = P^{-1} [\bar{\alpha}]_{\mathcal{B}}$$

$$\forall \bar{\alpha} \in V.$$

$$\left(\begin{aligned} \bar{\alpha} &= \sum_{j=1}^n x_j' \alpha_j' = \sum_{i=1}^n x_i \bar{\alpha}_i \\ \bar{\alpha}_j' &= \sum_{i=1}^n P_{ij} \bar{\alpha}_i \end{aligned} \right.$$

Example: $\mathbb{R}$ be real field & $\theta \in \mathbb{R}$ is fixed.
 $P = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, P^{-1} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$

$$\mathcal{B}' = \{ (\cos \theta, \sin \theta), (-\sin \theta, \cos \theta) \} \subset \mathbb{R}^2$$

$$\cancel{\mathcal{B} = \{ \bar{\alpha}_1, \bar{\alpha}_2 \}} [\bar{\alpha}]_{\mathcal{B}'} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$(X' = P^{-1} X)$$